1. **A new compound developed in the lab was tested as a mutagen. When the Ames test was performed, the results showed the presence of many bacterial colonies following exposure. This result means the chemical ______.**
    - a. causes apoptosis.
    - b. is radioactive.
    - c. would be a great candidate to use for chemotherapy.
    - d. is a mutagen and also may be a carcinogen.
    - e. is safe.
2. **The viral oncogene associated with the Rous sarcoma virus is the ______.**
    - a. ERBB2 gene.
    - b. v-sis gene.
    - c. src gene.
    - d. p53 gene.
    - e. RAS gene.
3. **Which of the following is a lipophilic primary message that crosses the plasma membrane and interacts with a cytosolic receptor?**
    - a. Epidermal Growth Factor
    - b. TGF-beta
    - c. Estrogen
    - d. Pertussis toxin
    - e. PDGF
4. **Each of the following components is associated with the primary cell wall except ______:**
    - a. lignins.
    - b. cellulose microfibrils.
    - c. hemicelluloses.
    - d. pectins.
    - e. glycoproteins.
5. **Each of the following is true about Ras except ______.**
    - a. Ras activity is inhibited by GTPase activating proteins (GAPs).
    - b. Ras activates a cascade of phosphorylation events within the cell.

- c. Ras binds steroid hormones.
- d. Ras is monomeric.
- e. Ras is activated by Sos, a guanine nucleotide exchange factor.

6. **Each of the following is true about basal lamina except ______.**
    - a. it is composed primarily of fibronectin.
    - b. it serves as a structural support.
    - c. it can influence cell migrations in certain areas of the body.
    - d. it allows for movement of some small molecules but blocks movement of large molecules.
7. **Crawling cells produce broad, thin protrusions known as ______ and pointed structures called ______ at the leading edge of the cell.**
    - a. filopodia; lamellipodia
    - b. podosome; lamellipodia
    - c. pseudopodia; focal contacts
    - d. lamellipodia; filopodia
8. **Actin bundles give shape and support to ______ found in intestinal mucosal cells.**
    - a. flagella
    - b. nuclei
    - c. microvilli
    - d. lamellipodia
    - e. cilia
9. **Compounds that inhibit receptors by preventing the natural messenger from binding are known as ______.**
    - a. antagonists.
    - b. agonists.
    - c. syndecans.
    - d. arrestins.
10. **The promotion phase of carcinogenesis is enhanced by ______.**
    - a. natural selection of cells exhibiting enhanced growth rate and invasive properties.
    - b. a single base-pair change in genomic DNA.

- c. inactivation of Ras activity.
- d. prolonged or repeated exposure to foreign or natural agents that stimulate cell proliferation.
- e. a single exposure to a carcinogen.

Answers for Questions 1-10:

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
D	C	C	A	C	A	D	C	A	D

Questions (Continued):

11. **Elastin molecules are crosslinked to each other by ______.**
    - a. peptide bonds between glycines.
    - b. hydrophobic interactions between hydroxyprolines and glycines.
    - c. none of these choices (above or below).
    - d. hydrogen bonds between hydroxylysines and hydroxyprolines.
    - e. covalent bonds between prolines.
12. **Cytoskeletal elements are found in ______.**
    - a. all of these choices (above and below).
    - b. archaea and eukaryotes.
    - c. eukaryotes.
    - d. bacteria.
    - e. archaea.
13. **Each of the following is true about plant pectins except ______.**
    - a. pectins trap and bind water molecules.
    - b. pectins are long unbranched polymers.
    - c. pectins form the matrix in which cellulose microfibrils are embedded.
    - d. pectins have a gel-like consistency.
    - e. pectins bind adjacent cell walls together.

14. **Each of the following is an intracellular messenger except ______.**
    - a. calcium ions.
    - b. cAMP.
    - c. PDGF.
    - d. DAG.
    - e. IP3.
15. **Each of the following is an example of microfilament-based motility and contractility except ______.**
    - a. cytoplasmic streaming.
    - b. cytokinesis.
    - c. trailing edge retraction in a moving cell.
    - d. ciliary motility.
    - e. pseudopodia formation.
16. **Children with hereditary retinoblastoma inherit a chromosome 13 from one parent that has which of the following defects in the RB gene?**
    - a. duplication
    - b. inversion
    - c. insertion
    - d. point mutation
    - e. deletion
17. **Transforming growth factor beta receptor is a ______.**
    - a. Receptor serine-threonine kinase
    - b. Non receptor serine/threonine kinase
    - c. G-protein coupled receptor
    - d. Receptor tyrosine kinase
18. **Which of the following would be correctly described as juxtacrine signaling?**
    - a. a signal that requires physical contact between cells
    - b. a signal that acts on the cell that produces it
    - c. a signal that diffuses over long distances from the producing cell

- d. a signal that diffuses over short distances from the producing cell
- e. a hormonal signal

19. **N-CAM allows interactions between neural cells and is a member of which of the following protein families?**
    - a. lectins
    - b. catenins
    - c. selectins
    - d. immunoglobulin superfamily (IgSF) proteins
    - e. cadherins
20. **Microfilaments function in ______.**
    - a. maintenance of animal cell shape.
    - b. cell division.
    - c. cytoplasmic streaming.
    - d. all of these (above and below).
    - e. muscle contraction.

Answers for Questions 11-20:

Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20
C	A	B	C	D	E	A	A	D	D

Questions (Continued):

21. **Most hereditary forms of breast and ovarian cancer arise in women who inherit a mutant copy of either the BRCA1 or BRCA2 gene, each of which encodes proteins involved directly in ______.**
    - a. inhibiting apoptosis.
    - b. a spindle assembly checkpoint.
    - c. repairing DNA double-strand breaks.
    - d. passing through the G1 restriction point.
    - e. activating Rho proteins to increase cell motility.

22. **Oncogenes encode proteins in each of the following categories except ______.**

- a. growth factors.
- b. nonreceptor protein kinases.
- c. cytochrome P450 enzymes.
- d. transcription factors.
- e. growth-factor receptors.

23. **In the axoneme, dynein sidearms are anchored on ______.**

- a. B tubules.
- b. interdoublet nexin.
- c. A tubules.
- d. central pair tubules.
- e. radial spokes.

24. **CapZ ______.**

- a. connects microfilaments to microtubules.
- b. causes microfilaments to lengthen by assembly at their plus ends.
- c. forms microfilament branches.
- d. depolymerizes microfilaments at their plus ends.
- e. prevents assembly at microfilament plus ends.

25. **Which of the following statements is correct with respect to flagella and cilia?**

- a. Cilia often have a helical pattern of movement.
- b. Cilia generally are longer than flagella.
- c. Cilia generate a force parallel to the cell surface.
- d. The flagellum is usually at the leading end of a swimming sperm cell.
- e. There usually are fewer cilia than flagella.

26. **Which of the following proteins attaches a crawling cell to the substrate in a focal adhesion?**

- a. integrin
- b. actin
- c. myosins
- d. alpha-actinin

27. **The ______ is a complex network of interconnected filaments and tubules that extends throughout the cytosol from the nucleus to the inner surface of the plasma membrane.**
    - a. cytoskeleton
    - b. desmin sheath
    - c. microfilament array
    - d. axoneme
    - e. microtubule apparatus
28. **Striated muscle has light and dark bands. The light bands are called _____, whereas the dark bands are called _____.**
    - a. H zones; M zones
    - b. Z lines; M lines
    - c. M lines; A bands
    - d. I bands; A bands
    - e. A bands; I bands
29. **In cartilage, proteoglycans form large complexes by attaching to ______.**
    - a. elastin.
    - b. glycosaminoglycans.
    - c. laminin.
    - d. collagen.
    - e. hyaluronate.
30. **How do carcinogens damage DNA?**
    - a. by causing breaks in one or both strands.
    - b. by generating cross-links between DNA strands.
    - c. all these choices (above and below) are mechanisms by which carcinogens can damage DNA.
    - d. by hydroxylating DNA bases.
    - e. by binding to DNA and directly disrupting base pairing.

Answers for Questions 21-30:

Q21	Q22	Q23	Q24	Q25	Q26	Q27	Q28	Q29	Q30
C	C	C	E	D	A	A	D	E	C

Questions (Continued):

31. **The expression of which of the following contributes to development of cancer?**
    - a. oncogenes
    - b. apoptosis genes
    - c. proto-oncogenes
    - d. tumor suppressor genes
    - e. the p53 gene
32. **Dynamic assembly and disassembly of a cytoplasmic microtubule in a cell occurs primarily at its plus end, because its minus end is usually anchored to ______.**
    - a. a centromere.
    - b. a microtubule organizing center.
    - c. the plasma membrane.
    - d. a mitochondrial outer membrane.
    - e. a basal body.
33. **IP3 receptors are associated with ______.**
    - a. mitochondrial membranes.
    - b. ER membranes.
    - c. endosome membranes.
    - d. plasma membranes.
    - e. Golgi apparatus membranes.
34. **cAMP activates ______.**
    - a. MAPK.
    - b. protein kinase A.
    - c. protein kinase G.
    - d. protein kinase B.
    - e. protein kinase C.

35. **Which of the following forms the insoluble layer of aromatic alcohols found primarily in woody tissues?**

    - a. proteoglycans
    - b. lignins
    - c. cadherins
    - d. pectins
    - e. hyaluronates

36. **Which of the following sequences is in the correct order for the process by which a myosin head generates force on an actin filament for muscle contraction?**

    - a. crossbridge formation, cocking of myosin head, power stroke, crossbridge dissociation
    - b. power stroke, crossbridge formation, crossbridge dissociation, cocking of myosin head
    - c. crossbridge formation, power stroke, crossbridge dissociation, cocking of myosin head
    - d. cocking of myosin head, crossbridge formation, power stroke, crossbridge dissociation

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Q31	**Q32**	**Q33**	**Q34**	**Q35**	**Q36**
A	B	B	B	B	C